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31 October 2026

Data 260 Homework 1

Personal Configuration and Domain

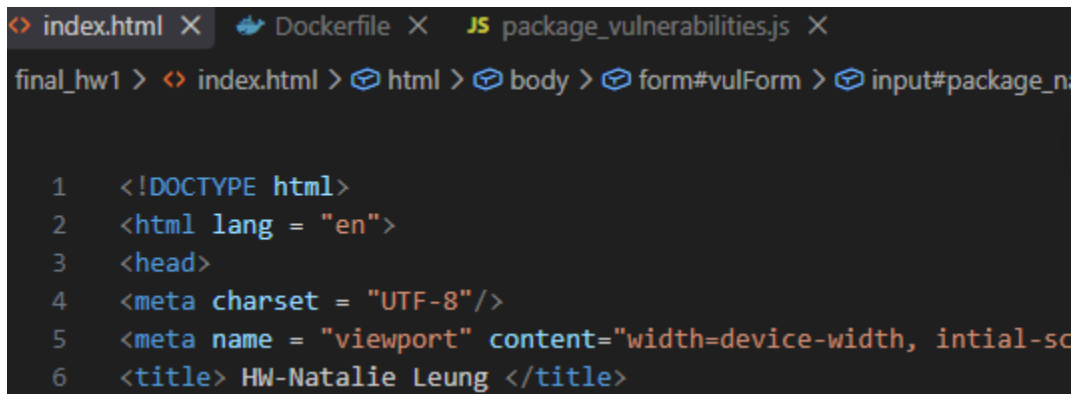
SID4	0932
PORT_BASE	8032
PREFIX	s0932
SEED	0932
VERIFY_SEED	260932
DOMAIN_ID	4 (open-source package vulnerabilities)

Hardware: nvidia geforce 4070, msi cpu
Local model used: ollama qwen3:8b
Tagged commit hash repo (submission on canvas only):

Questions: Part 1

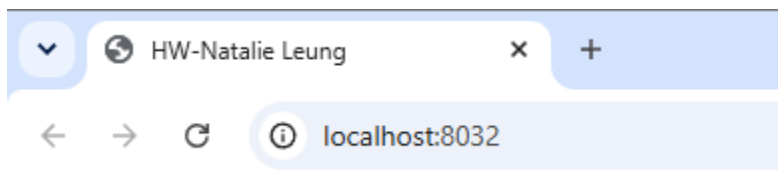
HTML

- A. Create an HTML page titled HW1-{Your name}



```
index.html X Dockerfile X JS package_vulnerabilities.js X
final_hw1 > index.html > html > body > form#vulForm > input#package_n

1  <!DOCTYPE html>
2  <html lang = "en">
3  <head>
4  <meta charset = "UTF-8"/>
5  <meta name = "viewport" content="width=device-width, initial-sc
6  <title> HW-Natalie Leung </title>
```



- B. Adding a header tag naming your assigned domain entity (use the header tag that renders the biggest font size)

localhost:8032

Open-Source Package Vulnerability Form

Package Name *

```
17 h1{
18   color: #333;
19   text-align: center;
20   margin-bottom: 30px;
21   margin-top: 10px;
22   border-bottom: 2px solid #4CAF50;
23 }
```

```
115 <body>
116   <h1>Open-Source Package Vulnerability Form</h1>
```

C. Create DOMAIN_SCHEMA.md

```
DOMAIN_SCHEMA.md # Vulnerability Form
1 # Vulnerability Form
2 | Field | Type | Required | Description |
3 |-----|-----|-----|-----|
4 | 'id' | string | not an input, generated with a Counter | generated with a Counter that keeps track of number of submissions |
5 | 'packageName' | text (auto focused, primary field) | yes | Full Open-Source Package Name (e.g Python npx module) |
6 | 'version' | text (secondary field) | yes | Version Number of the Package (e.g. 1.0.1) |
7 | 'email' | text (auto focused, primary field) | yes | an email of the user who submit the report like last.first@gmail.com |
8 | 'problem' | text (long input field) | yes | Describe the vulnerability, the risk it causes for the system, and/or if the open-source package version is EOL (Must be at least 25 characters long) |
9 | 'vulSeverity' | category (dropdown) | yes | severity level of the vulnerability threat |
10 | 'agreeTerms' | boolean (checkbox) | yes | I agree with the terms and conditions agreement |
11 | 'timestamp' | text (auto focused, primary field) | yes | time that the form was submitted |
12
13 # Severity Level
14 | Value | Meaning |
15 |-----|-----|
16 | 'Critical' | the open source package must be addressed within 24hrs to ensure that the package doesnt accidentally cause data leakage |
17 | 'High' | the vulnerability will do some damage or affect the code or work that uses the open source package at this very moment and should be fixed within 2-3 days |
18 | 'Medium' | might impact code or work that uses the open source package |
19 | 'Low' | minor issue, not a priority fix |
20
```

D. Create form for your domain entity

1. A required text input for the entity's primary field, with a placeholder describing it. The cursor should automatically focus on this field when the page loads.

Package Name *

Full Open-Source Package Name (e.g Python npx module)

```
<label for="package">Package Name *</label><br/>
<input type="text" id="package_name" placeholder="Full Open-Source Package Name (e.g Python npx module)" autofocus required /><br/><br/>
```

2. A required text input for the entity secondary field, with a descriptive placeholder

Version Number *

Version Number of the Package (e.g. 1.0.1)

```
<label for="version">Version Number *</label><br/>
<input type="text" id="version_num" placeholder="Version Number of the Package (e.g. 1.0.1)" required /><br/><br/>
```

3. A required email input for the submitter's email, with a descriptive placeholder

Email *

last.first@gmail.com

```
<label for="source">Email *</label><br/>
<input type="email" id="email_name" placeholder="last.first@gmail.com" required /><br/><br/>
```

4. A required text area

```
<label for="problem">Vulnerability Problem *</label><br/>
<textarea id="problem_note" rows="4" placeholder="Describe the vulnerability, the risk it causes for the system, and/or if the open-source package version is EOL (Must be at least 25 characters long)" required ></textarea><br/><br/>
```

Vulnerability Problem *

Describe the vulnerability, the risk it causes for the system, and/or if the open-source package version is EOL (Must be at least 25 characters long)

5. Dropdown category

Vulnerability Severity Level *

Low Risk

Choose
Critical Risk
High Risk
Medium Risk
Low Risk

```
<label for="vulSeverity">Vulnerability Severity Level *</label><br/>
  <select id="severity_level" required>
    <option value=""disabled selected>Choose</option>
    <option value="Critical Risk"selected>Critical Risk</option>
    <option value="High Risk"selected>High Risk</option>
    <option value="Medium Risk"selected>Medium Risk</option>
    <option value="Low Risk"selected>Low Risk</option>
  </select><br><br>
```

6. Check box

☐ I agree with the terms and conditions.

```
<div class="checkbox">
  <input type="checkbox" id="agreeTerms"/>
  <label for="agreeTerms">I agree with the terms and conditions.</label>
</div>
```

7. Submit button

Submit Vulnerability

```
<button type="submit">Submit Vulnerability</button>
```

E. Script tag to link your JavaScript code at end of your HTML file

```
148 <script src="package_vulnerabilities.js"></script>
```

Javascript

F. Write a JavaScript function using an arrow function to validate:

1. Verify that the content field has more than 25 characters. If validation fails, display an alert with an appropriate message

HW-Natalie Leung x +

localhost:8032

localhost:8032 says

Vulnerability description must be more than 25 characters long

OK

Open Vulnerability

Package Name *

Version Number *

Email *

Vulnerability Problem *

Vulnerability Severity Level *

Low Risk

☐ I agree with the terms and conditions.

Submit Vulnerability

Vulnerability Submissions

```
const problem = document.getElementById("problem_note").value.trim();

if(problem.length<=25){
  alert("Vulnerability description must be more than 25 characters long")
  return false;
}
```

2. Verify that the terms and conditions checkbox is needed. If validation fails, display an alert with an appropriate message

The screenshot shows a web browser window with the address bar displaying 'localhost:8032'. A modal alert box is open, titled 'localhost:8032 says', with the message 'Please agree with the terms and condition before submitting' and an 'OK' button. The background is a dark-themed form titled 'Vulnerability Submission'. The form contains the following fields:

- Package Name ***: Input field containing 'node.js'.
- Version Number ***: Input field containing '12.23.1'.
- Email ***: Input field containing 'zxcvzx@gmail.com'.
- Vulnerability Problem ***: Textarea containing the text: 'theres a vulnerability because this version has reached EOL and will be no longer supported after today'.
- Vulnerability Severity Level ***: Dropdown menu with 'High Risk' selected.
- Terms and Conditions**: A checkbox labeled 'I agree with the terms and conditions.' which is currently unchecked.
- Submit Vulnerability**: A green button.

Below the form is a section titled 'Vulnerability Submissions' with an empty input field.

```
const agreed = document.getElementById("agreeTerms").checked;

if(!agreed){
    alert("Please agree with the terms and condition before submitting")
    return false;
}
```

- G. After the form submission is successful, convert the form data into a JSON string and log the output in the console.

```
Package Vulnerabilities Data (String): package_vulnerabilities.js:61
{"packageName":"Node.js
Module","version":"22.12.3","email":"smith.john@gmail.com","problem":"there
s a vulnerability because this version has reached EOL and is no longer
supported","vulSeverity":"High Risk","timestamp":"2026-08-
31T00:20:35.704Z"}

Package Vulnerabilities Data (JSON): package_vulnerabilities.js:63
  {packageName: 'Node.js Module', version: '22.12.3', email: 'smith.john@gm
ail.com', problem: 'theres a vulnerability because this version has reach
ed EOL and is no longer supported', vulSeverity: 'High Risk', ...}
```

```
//json operations
const jsonvulData = JSON.stringify(vulData);
console.log("Package Vulnerabilities Data (String):", jsonvulData);
const parsedvulData = JSON.parse(jsonvulData);
console.log("Package Vulnerabilities Data (JSON):", parsedvulData);
```

- H. Use object destructuring to extract the primary field and email field from the parsed object and log their values in the console

```
Package Name: Node.js Module package_vulnerabilities.js:67
Version Number: 22.12.3 package_vulnerabilities.js:68
Email: smith.john@gmail.com package_vulnerabilities.js:69
Vulnerability Problem Explanation: theres a package_vulnerabilities.js:70
vulnerability because this version has reached EOL and is no longer
supported
Vulnerable Severity Level: High Risk package_vulnerabilities.js:71
```

```
//destructuring
const { packageName: package_name, version: version_num, email: email_name, problem: problem_note, vulSeverity: severity_level} = parsedvulData;
console.log("Package Name:", package_name);
console.log("Version Number:", version_num);
console.log("Email:", email_name);
console.log("Vulnerability Problem Explanation:", problem_note);
console.log("Vulnerable Severity Level:", severity_level);
```

- I. Spread operator

```
Updated Vuls: package_vulnerabilities.js:77
  {packageName: 'Node.js Module', version: '22.12.3', email: 'smith.john@gm
ail.com', problem: 'theres a vulnerability because this version has reach
ed EOL and is no longer supported', vulSeverity: 'High Risk', ...}
    email: "smith.john@gmail.com"
    id: "vul-1"
    packageName: "Node.js Module"
    problem: "theres a vulnerability because this version has reached EOL a
timestamp: "2026-08-31T00:20:35.704Z"
    version: "22.12.3"
    vulSeverity: "High Risk"
  } [[Prototype]]: Object
```

```
//spread operator and closures
const currentCount = vulCounter();
const updatedVul = { ...parsedvulData, id: `vul-${currentCount}`};
console.log("Vul Counter: ", currentCount);
console.log("Updated Vuls: ", updatedVul)
```

J. Closure to track times form submission

Vul Counter: 1 [package vulnerabilities.js:76](#)

```
//spread operator and closures
const currentCount = vulCounter();
const updatedVul = { ...parsedvulData, id: `vul-${currentCount}`};
console.log("Vul Counter: ", currentCount);
console.log("Updated Vuls: ", updatedVul)
```

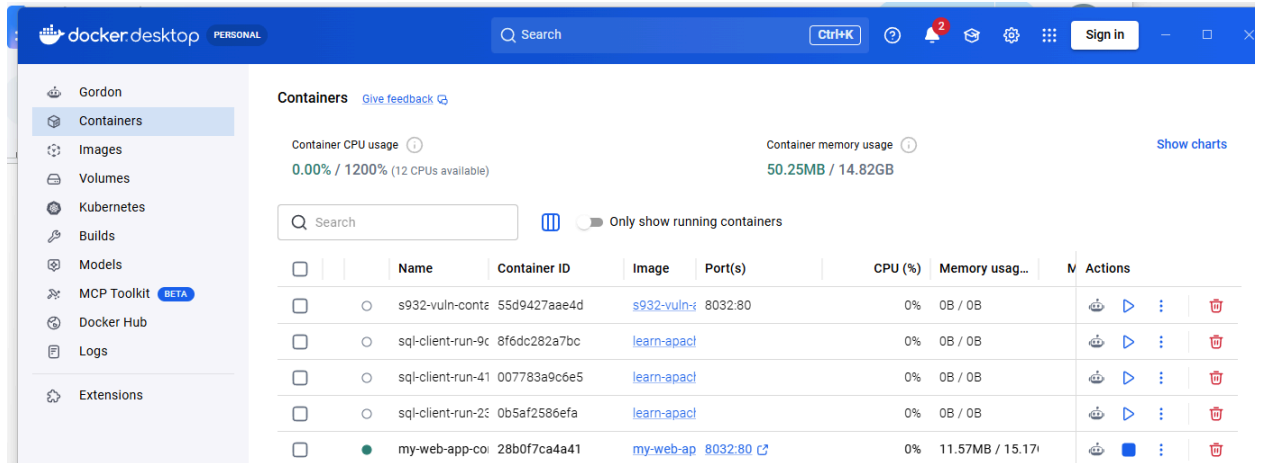
Deployment

K. Docker Screenshot

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\code> docker build -t my-web-app .

>> docker run -d -p 8032:80 --name my-web-app-container my-web-app
[+] Building 2.7s (8/8) FINISHED
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 187B 0.0s
=> [internal] load metadata for docker.io/library/nginx:alpine 2.1s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/3] FROM docker.io/library/nginx:alpine@sha256:db35bfc6b2951e7f8a72db5db120288c127ffaeeb4a6d4b95a26fead017d5913 0.0s
=> => resolve docker.io/library/nginx:alpine@sha256:db35bfc6b2951e7f8a72db5db120288c127ffaeeb4a6d4b95a26fead017d5913 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 364B 0.0s
=> CACHED [2/3] RUN rm -rf /usr/share/nginx/html/* 0.0s
=> [3/3] COPY . web_application/ /usr/share/nginx/html/ 0.1s
=> exporting to image 0.3s
=> => exporting layers 0.1s
=> => exporting manifest sha256:bc972b25565b0a699009f879f62672aa50cf71e92c34592acfb373c6ac290bd7 0.0s
=> => exporting config sha256:dbcc802a6a788e08719cb49d0688220e4f2cb3ced8dd2e71ee8820c7ecd7f3c6 0.0s
=> => exporting attestation manifest sha256:337f860ddb6b314df0b4391cc99912b3f09effd6231b58787d42031e835c7a34 0.0s
=> => exporting manifest list sha256:abdfb0a258215106183cab1f95a2c212c8310ef5aa6a271ed4a4484de7a9a4 0.0s
=> => naming to docker.io/library/my-web-app:latest 0.0s
=> => unpacking to docker.io/library/my-web-app:latest 0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/9vi99nyycb6mofuvydsibe5128b0f7ca4a41427e62509816d345d5e7c828426fbd3c27677539c8c78e8cff4
```



Open-Source Package Vulnerability Form

Package Name *

Full Open-Source Package Name (e.g Python npx module)

Version Number *

Version Number of the Package (e.g. 1.0.1)

Email *

last.first@gmail.com

Vulnerability Problem *

Describe the vulnerability, the risk it causes for the system, and/or if the open-source package version is EOL (Must be at least 25 characters long)

Vulnerability Severity Level *

Low Risk

☐ I agree with the terms and conditions.

Submit Vulnerability

Vulnerability Submissions

L. AWS ECS service (one task) where app running on my public IP address

```
PS C:\WINDOWS\system32> aws login
No AWS region has been configured. The AWS region is the geographic location of your AWS resources.

If you have used AWS before and already have resources in your account, specify which region they were created in. If you
have not created resources in your account before, you can pick the region closest to you: https://docs.aws.amazon.com
/global-infrastructure/latest/regions/aws-regions.html.

You are able to change the region in the CLI at any time with the command "aws configure set region NEW_REGION".
AWS Region [us-east-1]: us-east-1
Attempting to open your default browser. If the browser does not open, open the following URL.
If you are unable to open the URL on this device, run this command again with the '--remote' option.

https://us-east-1.signin.aws.amazon.com/v1/authorize?response_type=code&client_id=arn%3Aaws%3Asignin%3A%3A%3Adevtools%2F
same-device&state=a2575571-df91-4ed5-b61d-096e61e80bb5&code_challenge_method=SHA-256&scope=openid&redirect_uri=http%3A%2
F%2F127.0.0.1%3A65505%2Foauth%2Fcallback&code_challenge=NYy9JEqfTqTqamUBs7azUtQbd8W6-isCimSQHNeZVc

Updated profile default to use arn:aws:iam::872302839503:root credentials.
PS C:\WINDOWS\system32> aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 872
302839503.dkr.ecr.us-east-1.amazonaws.com
Login Succeeded

PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\code> docker tag my-web-app:latest 872302839503.dkr.ecr.us-east-1.a
amazonaws.com/my-web-app:latest
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\code> docker push 872302839503.dkr.ecr.us-east-1.amazonaws.com/my-w
eb-app:latest
The push refers to repository [872302839503.dkr.ecr.us-east-1.amazonaws.com/my-web-app]
c3ffca85b626: Pushed
44136fa355b3: Pushed
d94291c26261: Pushed
55afa1ecc21d: Pushed
0ea935727878: Pushed
0d62d88506ba: Pushed
1ed8e39a7434: Pushed
8cdf4f23778: Pushed
da4dea1b00af: Pushed
fb597529c916: Pushed
2ee81bd86624: Pushed
d33ebb2c248a: Pushed
latest: digest: sha256:abdffb0a258215106183cabcf95a2c212c8310ef5aa6a271ed4a4484de7a9a4 size: 856
```

Create security group

Security group (sg-0abf2bf437272c216 | my-web-app-sg) was created successfully

Details

sg-0abf2bf437272c216 - my-web-app-sg

Actions

Details

Security group name my-web-app-sg	Security group ID sg-0abf2bf437272c216	Description Security group for my web app	VPC ID vpc-042840cf09b113730
Owner 872302839503	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1)

Manage tags

Edit inbound rules

Search

	Name	Security group rule ID	IP version	Type	Protocol
☐	-	sgr-09bce15117adc04a4	IPv4	HTTP	TCP

Create clusters

Cluster my-web-app-cluster has been created successfully.
View cluster

Clusters (1) [Info](#)

By default, we only load up to 1,000 clusters at a time.

Cluster

Services

Tasks

Container instances

CloudWatch monitori

my-web-app-cluster	0	No tasks running	0 EC2	Default
------------------------------------	---	------------------	-------	---------

Task definition

Task definition successfully created
my-web-app-task:2 has been successfully created. You can use this task definition to deploy a service or run a task.
View task definition

my-web-app-task:2

Last updated
August 30, 2026, 22:11 (UTC-7:00)
Deploy
Actions
Create new revision

Overview [Info](#)

ARN
arn:aws:ecs:us-east-1:872302839503:task-definition/my-web-app-task:2

Status
ACTIVE

Time created
August 30, 2026, 22:11 (UTC-7:00)

App environment
Fargate

Task role
ecsTaskExecutionRole

Task execution role
ecsTaskExecutionRole

Operating system/Architecture
Linux/X86_64

Network mode
awsvpc

Fault injection
Turned off

Containers

JSON

Task placement

Volumes (0)

Requires attributes

Tags

Task size

Task CPU
256 units (0.25 vCPU)

Task CPU maximum allocation for containers

CPU (unit)
0 50 100 150 200 250
web-app-container Shared task CPU

Task memory
512 MiB (0.5 GiB)

Task memory maximum allocation for container memory reservation

Memory (MiB)
0 100 200 300 400 500
web-app-container Shared task memory

Container definition

Container: web-app-container

Info

Essential container

Details

JSON

Image

872302839503.dkr.ecr.us-east-1.amazonaws.com/my-web-app@sha256:abdfb0a258215106183cab1f95a2c212c8310ef5aa6a271ed4a4484de7a9a4

Private registry

Turned off

Secrets Manager ARN or name

-

CPU

0

Memory hard/soft limit

-/-

GPU

-

Neuron device

-

Environment and secrets

Network settings

Security and permissions

Lifecycle and dependencies

Environment variables (0)

Key

Type

Value

No environment variables

No environment variables to display.

Environment files (S3 ARN)

-

ECS Service

my-web-app-service has been deployed successfully.

View service

Notifications

0

0

3

0

0

my-web-app-cluster

Last updated

August 30, 2026, 22:14 (UTC-7:00)

Actions

Create with Express Mode

Cluster overview

Cluster ARN

arn:aws:ecs:us-east-1:87...

Status

Active

CloudWatch monitoring

Default

Action logs

Turned off

Services

0 active | 0 draining

Tasks

0 running | 0 pending

Registered container instances

-

Services

Tasks

Daemons

Infrastructure

Metrics

Scheduled tasks

Configuration - updated

Services (1)

Info

Last updated

August 30, 2026, 22:20 (UTC-7:00)

Manage tags

Update

Delete service

Create

Filter services by value

Filter launch type

Any launch type

Filter scheduling strategy

Any scheduling strategy

Filter resource management type

Any resource management type

1

Service name

ARN

Status

Schedu...

L...

Task de...

D

my-web-app-service

arn:aws:ec...

Active

Replica

-

my-web-a...

Deployment

my-web-app-service has been deployed successfully.

View service

Notifications00300

my-web-app-cluster

Last updatedAugust 30, 2026, 22:14 (UTC-7:00)

Actions

Create with Express Mode

Cluster overview

Cluster ARN
am:aws:ecs:us-east-1:87...

Status
Active

CloudWatch monitoring
Default

Action logs
Turned off

Services
0 active | 0 draining

Tasks
0 running | 0 pending

Registered container instances
-

Services

Tasks

Daemons

Infrastructure

Metrics

Scheduled tasks

Configuration - updated

Tasks (1)

Last updatedAugust 30, 2026, 22:21 (UTC-7:00)

Manage tags

Stop

Run new task

Filter tasks by property or value

Filter desired statusAny desired status

Filter launch typeAny launch type

< 1 >

Task	Last status	Desired status	Task definition	Health status
3470a5ac914d4945...	Running	Running	my-web-app-task:2	Unknown

Application running on public IP

Open-Source Package Vulnerability Form

Package Name *

D|

Version Number *

Version Number of the Package (e.g. 1.0.1)

Email *

last.first@gmail.com

Vulnerability Problem *

Describe the vulnerability, the risk it causes for the system, and/or if the open-source package version is EOL (Must be at least 25 characters long)

Vulnerability Severity Level *

Low Risk

☐ I agree with the terms and conditions.

Submit Vulnerability

Vulnerability Submissions

Agentic AI: Part 2

Deliverables

A. Command used

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\code> python .\agents_demo.py --title "Silicon VS Plastic Toys" --content "Many toys now lean away from plastic as silicon provides a fun new texture. Its flexible structure and softness are more appealing to parents as there are less risk of their child being hurt by these toys." --email "test@sjsu.edu" --strict
```

B. Output

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\code> python .\agents_demo.py --title "Silicon VS Plastic Toys" --content "Many toys now lean away from plastic as silicon provides a fun new texture. Its flexible structure and softness are more appealing to parents as there are less risk of their child being hurt by these toys." --email "test@sjsu.edu" --strict

--- Planner (16640 ms) ---
{
  "thought": "Identified key topics: material comparison, child safety, and texture appeal. Formulated relevant tags and a concise summary.",
  "message": "Tags and summary created based on the blog's focus on silicon vs. plastic toys, safety, and material properties.",
  "data": {
    "tags": [
      "silicon vs plastic toys",
      "child safety in toys",
      "flexible toy materials"
    ],
    "summary": "Silicon toys offer a safer, softer alternative to plastic, appealing to parents due to their flexibility and reduced injury risk.",
    "issues": []
  }
}

--- Planner (9820 ms) ---
{
  "thought": "Identified key topics: material comparison, child safety, and texture appeal. Formulated relevant tags and a concise summary.",
  "message": "Tags and summary created based on the blog's focus on silicon vs. plastic toys, safety, and material properties.",
  "data": {
    "tags": [
      "silicon vs plastic toys",
      "child safety in toys",
      "flexible toy materials"
    ],
    "summary": "Silicon toys offer a safer, softer alternative to plastic, appealing to parents due to their flexibility and reduced injury risk.",
    "issues": []
  }
}

Finalized Output
{
  "thought": "Finalizing tags and summary based on material comparison, safety focus, and tactile properties emphasized in the blog.",
  "message": "Tags and summary reflect the blog's emphasis on silicone's safety advantages and material characteristics over plastic toys.",
  "data": {
    "tags": [
      "silicone vs plastic toys",
      "toy safety materials",
      "flexible toy materials"
    ],
    "summary": "Silicone toys offer safer, softer alternatives to plastic, appealing to parents due to reduced injury risks.",
    "issues": []
  }
}
```

```

Publish Package
{
  "title": "Silicon VS Plastic Toys",
  "email": "test@sjsu.edu",
  "content": "Many toys now lean away from plastic as silicon provides a fun new texture. Its flexible structure and softness are more appealing to parents as there are less risk of their child being hurt by these toys.",
  "agents": {
    "transcript": [
      {
        "role": "Planner",
        "content": "Tags and summary created based on the blog's focus on silicon vs. plastic toys, safety, and material properties."
      },
      {
        "role": "Reviewer",
        "content": "Tags and summary created based on the blog's focus on silicon vs. plastic toys, safety, and material properties."
      }
    ],
    "final": {
      "tags": [
        "silicone vs plastic toys",
        "toy safety materials",
        "flexible toy materials"
      ],
      "summary": "Silicone toys offer safer, softer alternatives to plastic, appealing to parents due to reduced injury risks."
    },
    "issues": []
  }
},
"submissionDate": "2026-08-d07:23:4507:23:45Z"
}

```

C. Questions

1. The 3 final tags are "silicone vs plastic toys", "toy safety materials", and "flexible toy materials".
2. The final summary that is less or equal to 25 words is "Silicone toys offer safer, softer alternatives to plastic, appealing to parents due to reduced injury risks."
3. The tags were changed where, for example, "child safety in toys" was changed to "toy safety materials." The summary was slightly reworded, but the reviewer agent didn't change anything as the output in the console shows that the reviewer's input is the same as the planner's output and the changes only are shown in the finalized output.

D. First, the planner reads the raw data that includes the title and content of the blog. It then makes a tag set and summary that's less than 25 words long. This output is just based on the input itself. Then, the reviewer takes the planner's output and reevaluates it and uses that as the input. The reviewer's output is then used for the finalizer and produces a finalized output, which is a different output for the tags and summary compared to the planner's output. The finalized tags and summary are then put into the publish package along with the original blog input content.

Measuring Non-Determinism: Part 3

- I. Run agents_demo.py pipeline on exact input 20 times at temp 0.7 and 20 times at temp 0.0 (40 runs total)

```

r temp in [0.7,0.0]:
    label=str(temp).replace(".", "_")
    runs=[]
    for i in range(1,21):
        print(f"temp={temp} run={i}/20", file=sys.stderr)
        output, tags, latency = run(case["title"], case["content"], case["email"], temp)

        with open(os.path.join(outputFile, f"run_temp{label}_{i:02d}.txt"), "w") as f:
            f.write(output)
        runs.append({"run":i, "tags": tags, "latency_ms":latency})
# save every run as a json and csv
with open(os.path.join(outputFile, f"nondeterminism_temp_{label}.json"), "w") as f:
    json.dump(runs, f, indent=2)
with open(os.path.join(outputFile, f"nondeterminism_temp_{label}.csv"), "w", newline="") as f:
    w= csv.writer(f)
    w.writerow(["run", "tag_1", "tag_2", "tag_3", "latency_ms"])
    for r in runs:
        tags=(r["tags"]+["", "", ""])[0:3]
        w.writerow([r["run"], *tags, r["latency_ms"]])
summary[label] = get_metrics(runs)

```

```

PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\code> python .\nondeterminism.py
temp=0.7 run=1/20
temp=0.7 run=2/20
temp=0.7 run=3/20
temp=0.7 run=4/20
temp=0.7 run=5/20
temp=0.7 run=6/20
temp=0.7 run=7/20
temp=0.7 run=8/20
temp=0.7 run=9/20
temp=0.7 run=10/20
temp=0.7 run=11/20
temp=0.7 run=12/20
temp=0.7 run=13/20
temp=0.7 run=14/20
temp=0.7 run=15/20
temp=0.7 run=16/20
temp=0.7 run=17/20
temp=0.7 run=18/20
temp=0.7 run=19/20
temp=0.7 run=20/20
temp=0.0 run=1/20
temp=0.0 run=2/20
temp=0.0 run=3/20
temp=0.0 run=4/20
temp=0.0 run=5/20
temp=0.0 run=6/20
temp=0.0 run=7/20
temp=0.0 run=8/20
temp=0.0 run=9/20
temp=0.0 run=10/20
temp=0.0 run=11/20
temp=0.0 run=12/20
temp=0.0 run=13/20
temp=0.0 run=14/20
temp=0.0 run=15/20
temp=0.0 run=16/20
temp=0.0 run=17/20
temp=0.0 run=18/20
temp=0.0 run=19/20
temp=0.0 run=20/20

```

II. Each temperature report

```

# save every run as a json and csv
with open(os.path.join(outputFile, f"nondeterminism_temp_{label}.json"), "w") as f:
    json.dump(runs, f, indent=2)
with open(os.path.join(outputFile, f"nondeterminism_temp_{label}.csv"), "w", newline="") as f:
    w = csv.writer(f)
    w.writerow(["run", "tag_1", "tag_2", "tag_3", "latency_ms"])
    for r in runs:
        tags = (r["tags"] + ["", "", ""])[0:3]
        w.writerow([r["run"], *tags, r["latency_ms"]])

```

```

PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\code> python .\nondeterminism.py
temp=0.7 run=1/20
temp=0.7 run=2/20
temp=0.7 run=3/20

```

```

Metrics table
'Metric' | 'Temp 0.7' | 'Temp 0.0'
Distinct tag sets | 17 | 2
Tags in all 20 runs | [] | ['silicone toys']
Tags in exactly 1 run | ['child safety', 'child safety in toys', 'child safety toys', 'flexible toy materials', 'material innovation', 'parent appeal', 'parent preference', 'parent preferences', 'plastic vs silicone', 'toy safety benefits', 'sensory play', 'silicon toys safety', 'silicone materials', 'silicone texture', 'silicone toy safety', 'silicone toys safety', 'silicone vs plastic', 'silicone vs plastic toys', 'tactile appeal', 'tactile design', 'toy safety comparison', 'toys material innovation'] | ['plastic alternatives', 'toy safety']
Latency p80 (ms) | 26730.5 | 27073.0
Latency p95 (ms) | 35959.5 | 27411.45
Latency p99 (ms) | 45343.9 | 27454.29
Done

```

III. Save raw per-run results

```

# save every run as a json and csv
with open(os.path.join(outputFile, f"nondeterminism_temp_{label}.json"), "w") as f:
    json.dump(runs, f, indent=2)
with open(os.path.join(outputFile, f"nondeterminism_temp_{label}.csv"), "w", newline="") as f:
    w= csv.writer(f)
    w.writerow(["run", "tag_1", "tag_2", "tag_3", "latency_ms"])
    for r in runs:
        tags=(r["tags"]+["", "", ""])[ :3]
        w.writerow([r["run"], *tags, r["latency_ms"]])

```

```

reports > hw01 > raw > nondeterminism_temp_0_0.csv > data
1  run,tag_1,tag_2,tag_3,latency_ms
2  1,silicone toys,plastic alternatives,toy safety,27153
3  2,toys safety,silicone toys,plastic vs silicone,27061
4  3,toys safety,silicone toys,plastic vs silicone,26995
5  4,toys safety,silicone toys,plastic vs silicone,27195
6  5,toys safety,silicone toys,plastic vs silicone,27056
7  6,toys safety,silicone toys,plastic vs silicone,27041
8  7,toys safety,silicone toys,plastic vs silicone,27278
9  8,toys safety,silicone toys,plastic vs silicone,27016
10 9,toys safety,silicone toys,plastic vs silicone,27129
11 10,toys safety,silicone toys,plastic vs silicone,26902
12 11,toys safety,silicone toys,plastic vs silicone,26927
13 12,toys safety,silicone toys,plastic vs silicone,26839
14 13,toys safety,silicone toys,plastic vs silicone,26903
15 14,toys safety,silicone toys,plastic vs silicone,27102
16 15,toys safety,silicone toys,plastic vs silicone,26686
17 16,toys safety,silicone toys,plastic vs silicone,27363
18 17,toys safety,silicone toys,plastic vs silicone,27109
19 18,toys safety,silicone toys,plastic vs silicone,27211
20 19,toys safety,silicone toys,plastic vs silicone,27414
21 20,toys safety,silicone toys,plastic vs silicone,27095
22

```



```
client=ModelClient(model=args.model, base_url=args.base_url)
messages=[{"role":"system", "content":load()}]
```

```
src > model_client.py > ...
1  import ollama
2
3  class ModelClient:
4      def __init__(self, model="qwen3:8b",base_url="https://localhost:11434"):
5          self.model = model
6          self.client=ollama.Client(host=base_url)
7      def complete(self,messages,tools=None):
8          response=self.client.chat(model=self.model, messages=messages,tools=tools)
9          input_tokens=response.get("prompt_eval_count",0)
10         output_tokens=response.get("eval_count",0)
11         return{
12             "content":response["message"]["content"],
13             "input_tokens":input_tokens,
14             "output_tokens":output_tokens,
15             "total_tokens":input_tokens+output_tokens,
16         }
17
```

II. Hw1_client.py creation

```

code > hwt_client.py > ...
1  import argparse,json,os,sys
2
3  # have to do this so it gets the model_client.py bc located in differnt location
4  sys.path.insert(0,os.path.join(os.path.dirname(__file__),"..","src"))
5  from model_client import ModelClient
6  def load():
7      agent=os.path.join(os.path.dirname(__file__),"..","AGENT.md")
8      with open(agent)as f:
9          return f.read()
10 def stats(num_turns, total_in_tok, total_out_tok,messages):
11     l = len(json.dumps(messages))
12     print(f"\n[/stats] turns={num_turns} total_input_tokens={total_in_tok} total_output_tokens={total_out_tok} chars={l}\n")
13
14 def main():
15     ap=argparse.ArgumentParser()
16     ap.add_argument("--model",default=os.environ.get("SMOL_MODEL", "qwen3:8b"))
17     ap.add_argument("--base_url",default=os.environ.get("OLLAMA_URL", "http://localhost:11434"))
18     args=ap.parse_args()
19
20     client=ModelClient(model=args.model, base_url=args.base_url)
21     messages=[{"role":"system", "content":load()}]
22
23     num_turns=0
24     total_in_toks=0
25     total_out_toks=0
26
27     print("Type code to review, /stats for stats, or /exit to stop the program")
28     while True:
29         try:
30             user_input=input("you> ").strip()
31         except EOFError:
32             print("\n No input -- break")
33             break
34         if not user_input:
35             continue
36         if user_input==" /exit":
37             print(f"\nEnding stats: turns={num_turns} | total_in_tokens={total_in_toks} | total_out_tokens={total_out_toks}")
38             break
39         if user_input==" /stats":
40             stats(num_turns,total_in_toks,total_out_toks, messages)
41             continue
42
43         messages.append({"role":"user","content":user_input})
44         result=client.complete(messages)
45         messages.append({"role":"assistant","content":result["content"]})
46         num_turns+=1 #increment
47         total_in_toks+=result["input_tokens"]
48         total_out_toks+=result["output_tokens"]
49
50         print(f"assistant> {result['content']}")
51         print(f"turn {num_turns} | input_tokens={result['input_tokens']} | output_tokens={result['output_tokens']} | totak_tokens={result['total_tokens']}")
52
53 if __name__ == "__main__":
54     main()

```

```

PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\code> python .\hw1_client.py
Type code to review, /stats for stats, or /exit to stop the program
you> def divide(a,b):  if b=0:      x=0
assistant> - Syntax error: Incorrect use of `=` instead of `==` in the condition `if b=0:`
- NameError: Use of undefined variable `think` in the division `0 /think`
[turn 1] | input_tokens=128 | output_tokens=728 | total_tokens=856
you> else:      x=a/b
assistant> - Syntax error: Incorrect use of `=` instead of `==` in the condition `if b=0:`
- Syntax error: Missing colon `:` after `else` statement
- Syntax error: Incorrect indentation for the `else` block
- NameError: Use of undefined variable `think` in the division `0 /think`
- Logic error: Division by zero not properly handled in the function
[turn 2] | input_tokens=187 | output_tokens=445 | total_tokens=632
you> def divide(a,b):  if b==0:      x=0
assistant> - Syntax error: Invalid syntax in `x=0 /think` (unexpected ` /think`)
- Logic error: Missing `else` block to handle non-zero division
- Logic error: Function does not return the computed value `x`
- Syntax error: Missing colon `:` after `if` statement
- Syntax error: Incorrect indentation for the assignment `x=0`
[turn 3] | input_tokens=296 | output_tokens=375 | total_tokens=671
you> /stats

[/stats] turns=3 total_input_tokens=611 total_output_tokens=1548 chars=1615

you> def divide(a,b):  if b==0:      x=0 else:      x=a/b
assistant> - Syntax error: Unexpected ` /think` in expression `x=a/b /think`
- Syntax error: Missing colon `:` after `if` statement
- Syntax error: Missing colon `:` after `else` statement
- Syntax error: Incorrect indentation for the `else` block
- Logic error: Division by zero not properly handled in the function
- Syntax error: Incorrect indentation for the assignment `x=0`
[turn 4] | input_tokens=405 | output_tokens=2708 | total_tokens=3113
you> def divide(a,b):  if b==0:      x=0 return x else:      x=a/b return x
assistant> - Syntax error: Multiple statements on one line without semicolons (`x=0 return x`)
- Syntax error: Invalid token ` /think` at the end of the line
- Syntax error: Missing colon `:` after `if` and `else` statements
- Syntax error: Incorrect indentation for return statements
- Logic error: Return statements are on the same line as assignments, which is invalid syntax
- Syntax error: Missing colon `:` after `else` statement
- Logic error: Function does not handle division by zero properly (return statement is unreachable in `if` block)
[turn 5] | input_tokens=527 | output_tokens=857 | total_tokens=1384
you> /stats

[/stats] turns=5 total_input_tokens=1543 total_output_tokens=5113 chars=2854

you> /exit

```

III. [AGENT.md](#) follow bullet point

```

📌 AGENT.md
1  You are a code reviewer. Follow the following rules/guidelines for every response:
2  - Answer only in bullet points
3  - Do not include any introductions, or summaries, or paragraphs
4  - Each bullet point will only show one issue
5  - Do not repeat bullet point points, do not repeat any code to user
6  - Do not show any code to user
7  - If there are no issue, reply with one bullet point telling user "There are no errors in the code"

```

```

agent_ok, agent_detail=False, "file missing"
if file_exists("AGENT.md"):
    text=open(os.path.join(root, "AGENT.md")).read().lower()
    agent_ok="bullet" in text
    agent_detail="mentions bullet point format" if agent_ok else "did not mention bullet point formatting"
check("AGENT.md needs to say bullet point formating", agent_ok, agent_detail)

PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\code> python .\verify_hw01.py
{
  "timestamp": "2026-08-31T17:37:56.868631+00:00",
  "overall_stat": "PASS",
  "checks": [
    {
      "check": "AGENT.md exists at github repo",
      "passed": true,
      "detail": ""
    },
    {
      "check": "src/model_client.py exists at github repo",
      "passed": true,
      "detail": ""
    },
    {
      "check": "code/hw1_client.py exists at github repo",
      "passed": true,
      "detail": ""
    },
    {
      "check": "code/agents_demo.py exists at github repo",
      "passed": true,
      "detail": ""
    },
    {
      "check": "code/web_application/index.html exists at github repo",
      "passed": true,
      "detail": ""
    },
    {
      "check": "code/web_application/package_vulnerabilities.js exists at github repo",
      "passed": true,
      "detail": ""
    },
    {
      "check": "AGENT.md needs to say bullet point formating",
      "passed": true,
      "detail": "mentions bullet point format"
    },
    {
      "check": "Ollama server if connected or not",
      "passed": true,
      "detail": "Ollama is open and can be connected to"
    }
  ]
}
done

```

IV. Exit outputs the turns, and tokens

```

print(f"[turn {num_turns}] | input_tokens={result['input_tokens']} | output_tokens={result['output_tokens']} | total_tokens={result['total_tokens']}")

if user_input=="exit":
    print(f"\nEnding stats: turns={num_turns} | total_in_tokens={total_in_toks} | total_out_tokens={total_out_toks}")
    break

PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\code> python .\hw1_client.py
Type code to review, /stats for stats, or /exit to stop the program
you> def divide(a,b):  if b=0:      x=0
assistant> - Syntax error: Incorrect use of `=` instead of `==` in the condition `if b=0:`
- NameError: Use of undefined variable `think` in the division `0 /think`
[turn 1] | input_tokens=128 | output_tokens=728 | total_tokens=856
you> else:      x=a/b
assistant> - Syntax error: Incorrect use of `=` instead of `==` in the condition `if b=0:`
- Syntax error: Missing colon `:` after `else` statement
- Syntax error: Incorrect indentation for the `else` block
- NameError: Use of undefined variable `think` in the division `0 /think`
- Logic error: Division by zero not properly handled in the function
[turn 2] | input_tokens=187 | output_tokens=445 | total_tokens=632
you> def divide(a,b):  if b==0:      x=0
assistant> - Syntax error: Invalid syntax in `x=0 /think` (unexpected `/think`)
- Logic error: Missing `else` block to handle non-zero division
- Logic error: Function does not return the computed value `x`
- Syntax error: Missing colon `:` after `if` statement
- Syntax error: Incorrect indentation for the assignment `x=0`
[turn 3] | input_tokens=296 | output_tokens=375 | total_tokens=671
you> /stats

[/stats] turns=3 total_input_tokens=611 total_output_tokens=1548 chars=1615

you> def divide(a,b):  if b==0:      x=0 else:      x=a/b
assistant> - Syntax error: Unexpected `/think` in expression `x=a/b /think`
- Syntax error: Missing colon `:` after `if` statement
- Syntax error: Missing colon `:` after `else` statement
- Syntax error: Incorrect indentation for the `else` block
- Logic error: Division by zero not properly handled in the function
- Syntax error: Incorrect indentation for the assignment `x=0`
[turn 4] | input_tokens=405 | output_tokens=2708 | total_tokens=3113
you> def divide(a,b):  if b==0:      x=0 return x else:      x=a/b return x
assistant> - Syntax error: Multiple statements on one line without semicolons (`x=0 return x`)
- Syntax error: Invalid token `/think` at the end of the line
- Syntax error: Missing colon `:` after `if` and `else` statements
- Syntax error: Incorrect indentation for return statements
- Logic error: Return statements are on the same line as assignments, which is invalid syntax
- Syntax error: Missing colon `:` after `else` statement
- Logic error: Function does not handle division by zero properly (return statement is unreachable in `if` block)
[turn 5] | input_tokens=527 | output_tokens=857 | total_tokens=1384
you> /stats

[/stats] turns=5 total_input_tokens=1543 total_output_tokens=5113 chars=2854

you> /exit

Ending stats: turns=5 | total_in_tokens=1543 | total_out_tokens=5113

```

V. Add a stats command

```

def stats(num_turns, total_in_tok, total_out_tok, messages):
    l = len(json.dumps(messages))
    print(f"\n[/stats] turns={num_turns} total_input_tokens={total_in_tok} total_output_tokens={total_out_tok} chars={l}\n")

if user_input=="/stats":
    stats(num_turns, total_in_toks, total_out_toks, messages)
    continue

```

```
you> /stats

[/stats] turns=3 total_input_tokens=611 total_output_tokens=1548 chars=1615
```

VI. Run a five-turn convo

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\code> python .\hw1_client.py
Ending stats: turns=5 | total_in_tokens=0 | total_out_tokens=3550
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\code> python .\hw1_client.py
Type code to review, /stats for stats, or /exit to stop the program
you> def divide(a,b):  if b=0:      x=0
assistant> - Syntax error: Incorrect use of `=` instead of `==` in the condition `if b=0:`
- NameError: Use of undefined variable `think` in the division `0 /think`
[turn 1] | input_tokens=128 | output_tokens=728 | total_tokens=856
you> else:      x=a/b
assistant> - Syntax error: Incorrect use of `=` instead of `==` in the condition `if b=0:`
- Syntax error: Missing colon `:` after `else` statement
- Syntax error: Incorrect indentation for the `else` block
- NameError: Use of undefined variable `think` in the division `0 /think`
- Logic error: Division by zero not properly handled in the function
[turn 2] | input_tokens=187 | output_tokens=445 | total_tokens=632
you> def divide(a,b):  if b==0:      x=0
assistant> - Syntax error: Invalid syntax in `x=0 /think` (unexpected `/think`)
- Logic error: Missing `else` block to handle non-zero division
- Logic error: Function does not return the computed value `x`
- Syntax error: Missing colon `:` after `if` statement
- Syntax error: Incorrect indentation for the assignment `x=0`
[turn 3] | input_tokens=296 | output_tokens=375 | total_tokens=671
you> /stats

[/stats] turns=3 total_input_tokens=611 total_output_tokens=1548 chars=1615

you> def divide(a,b):  if b==0:      x=0 else:      x=a/b
assistant> - Syntax error: Unexpected `/think` in expression `x=a/b /think`
- Syntax error: Missing colon `:` after `if` statement
- Syntax error: Missing colon `:` after `else` statement
- Syntax error: Incorrect indentation for the `else` block
- Logic error: Division by zero not properly handled in the function
- Syntax error: Incorrect indentation for the assignment `x=0`
[turn 4] | input_tokens=405 | output_tokens=2708 | total_tokens=3113
you> def divide(a,b):  if b==0:      x=0 return x else:      x=a/b return x
assistant> - Syntax error: Multiple statements on one line without semicolons (`x=0 return x`)
- Syntax error: Invalid token `/think` at the end of the line
- Syntax error: Missing colon `:` after `if` and `else` statements
- Syntax error: Incorrect indentation for return statements
- Logic error: Return statements are on the same line as assignments, which is invalid syntax
- Syntax error: Missing colon `:` after `else` statement
- Logic error: Function does not handle division by zero properly (return statement is unreachable in `if` block)
[turn 5] | input_tokens=527 | output_tokens=857 | total_tokens=1384
you> /stats

[/stats] turns=5 total_input_tokens=1543 total_output_tokens=5113 chars=2854

you> /exit

Ending stats: turns=5 | total_in_tokens=1543 | total_out_tokens=5113 ^C
```

VII. Answer questions in Report and README

- A. The prior context is resent every term because the model doesn't have memory between the API calls made. Therefore, you need the client to send the previous conversation every time in order for the model to remember and go off on that.

- B. A system prompt is where to set rules and the roles for the models in the conversation. This affects how the interactions are done and affects everything. This is unchanged. A user message is a user input that affects it for one term and this is changed within the model as it summarizes it to be something else. This doesn't continue to carry importance later unlike the system prompt.
- C. Input tokens grow because each request causes a resending of the entire history for memory. It continues to add up as the conversation gets longer as it appends the history to the new message.
- D. The limits of growth are the model's max token limit. Because the input tokens add up quickly, the model can fail if the max token is reached when getting the resent memory. Tokens are also expensive and limited. It is difficult to add to the model's max token limit because this is another added cost.